

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Prepared and documented validation reports for **derivative valuation models**, **risk measurement models**, and **behavioural models**, presenting findings to stakeholders.
- Provided quantitative support to the wider Financial Risk team by developing analytical tools and performing ad-hoc data analysis using **Python** and **SQL**.
- Collaborated with model developers and key stakeholders to track and drive validation actions to completion, ensuring timely resolution of identified issues.
- Developed and maintained **Python** scripts to automate data extraction and transformation for risk analysis, processing approximately **1.5TB** of financial data weekly.
- Implemented and supported **Java Spring Boot** REST APIs for cross-system data exchange, ensuring robust communication between financial services platforms.
- Managed production environments for enterprise applications, responding to P1/P2 incidents and ensuring adherence to strict SLAs within a **GDPR-aware** operational framework.

SKILLS

Risk Modelling – Financial Risk Models, Derivative Valuation Models, Risk Measurement Models, Behavioural Models, Quantitative Analysis, Mathematical Modelling

Programming Languages – Python, R, Java, SQL

Financial Software – Calypso, FINCAD, QRM

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Warehousing, Pandas, NumPy

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Enterprise Systems – ERP Integration, REST APIs, Change Management, Agile/Scrum

PROJECTS

Financial Risk Model Validation Framework

- Developed a comprehensive framework in **Python** for validating financial risk models, incorporating automated testing for key parameters and assumptions.
- Engineered data pipelines using **Pandas** and **SQL** to prepare diverse datasets for model input, handling complex transaction data from multiple sources.
- Documented validation procedures and generated summary reports, detailing model performance against predefined metrics and regulatory requirements.

Automated Transaction Reconciliation System (HCL)

- Built an automated reconciliation system using **Java Spring Boot** and **PostgreSQL** to identify discrepancies in high-volume financial transaction data, reducing manual effort by an estimated **40%**.
- Implemented complex SQL queries and stored procedures to compare transaction records across different systems, flagging reconciliation failures for investigation.
- Integrated the system with an existing operational dashboard, providing real-time alerts and detailed logs for **transaction monitoring** and analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)